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Pressure Strength of Taped-On Doors

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Introduction

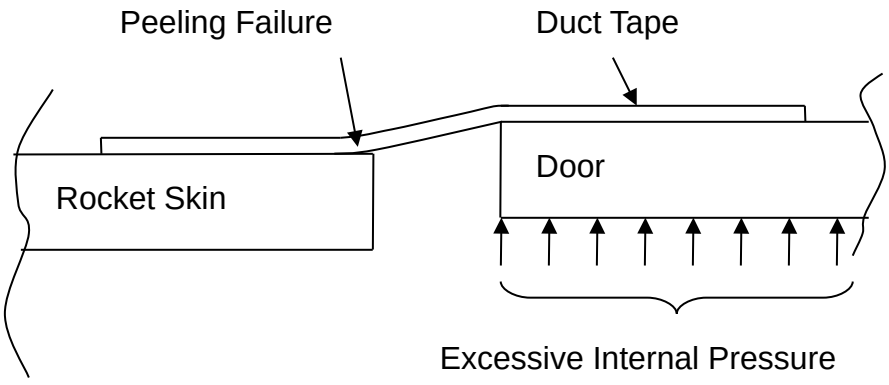
For three years running, the CSULB entry in the 10,000 ft ESRA competition, Gold Rush, flew to apogees far below predicted. Why? We weighed the rocket before flight, and used experimental thrust data for the motors. It had to be the drag. But, we used standard analysis techniques to estimate the drag. What could be happening?

Then, last year while awaiting the ESRA awards ceremony, Some folks from another university walked in with a piece of curved plexiglas. "Did anyone recognize this?" they asked. Yes, I did. It was the viewing window for our camera payload. "Where was it found?" "About 500 ft from our pad." Since the rocket was recovered about 1.5 miles down range from our pad, it was clear that the window had separated very early in flight, probably near burnout. So the scales fell from our eyes. The extra drag came from the very dirty compartments exposed when their doors blew off.

It had been our practice to attach doors (and windows) with duct tape, or something similar, around their edges. At the same time we'd been very casual about venting the underlying compartments during ascent, even though we had a perfectly good venting code on hand.

Analysis

Here, a simple model for door and window attachment failure will be developed. Under internal pressure a door will attempt to peel off the tape around its edges.



The sketch above illustrates the failure mechanism.

Now, note that the total expulsive force acting on a door $= \Delta p A_{door}$, where
Δp = The pressure difference, psi, between the internal compartment and the outside environment, and
A_{door} =The door area, in^2.
The maximum restraining force acting on the door $= \sigma_{peel} L$, where
σ_{peel} = The maximum peel strength provided by the duct tape-surface combination, lb/in, and L = The peripheral distance around the door, in

Thus failure (door expulsion) will occur when

$$\Delta p A_{door} \geq \sigma_{peel} L \text{ ., or}$$

$$\Delta p \frac{A_{door}}{L} \leq \sigma_{peel} .$$

Conclusions

Several observations are relevant. First, door size matters. Smaller doors are easier to retain than larger ones. Door shape also matters. A circular door maximizes A_{door} / L, and is therefore the worst choice. High aspect ratio rectangular doors are better than those of lower aspect ratio. A neat dodge is to crenellate or serrate the door edges. This will increase the door periphery without changing its area.

Finally, σ_{peel} must be found experimentally. It will depend on the kind of tape used and the surface preparation to which the tape adheres. A good experiment will also explore the geometric considerations described above.